METRIC

A-A-50624
9 June 1994

COMMERCIAL ITEM DESCRIPTION

AIRFLOW MONITORING EQUIPMENT

The General Services Administration has authorized the use of this commercial item description in preference to MIL-S-16032M.

Abstract.

This Commercial Item Description covers requirements for airflow monitoring equipment that measures and provides a display of airflow velocity, and generates alarms whenever airflow velocity drops below a predetermined level. Integral components include an airflow sensor and an associated airflow display panel. The equipments are intended for use in virtually any air ventilation system including those which service hazardous material stowage compartments where impeded airflow could result in dangerous conditions.

Salient characteristics.

1. Airflow monitoring system. The detailed requirements for airflow sensors and airflow display panels built in accordance with this CID are described in sections (2) and (3) below respectively. The basic performance requirements of an airflow monitoring system comprised of an airflow sensor and airflow display panel built in accordance with this CID are as follows:

a. Range. The airflow system shall accommodate selectable subranges within an overall airflow range of from 0 to 50 meters per second (mps).

b. Accuracy. Airflow velocity measurement accuracy shall be within ± (3.0 percent of reading plus 2.0 percent of full-scale). Low-airflow alarm point detection accuracy shall be within: ± (6.0 percent of the selected low-airflow point plus 1.0 percent full-scale).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160.

AMSC N/A FSC 6680

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c. Response time. When operated in a standard one-atmosphere ambient environment with an airflow velocity of 25 percent of full-scale, the airflow monitoring system shall achieve a full-accuracy output signal and full-accuracy display readout within 12 seconds for an airflow velocity step increase of from 30 to 75 percent full-scale.

d. Repeatability. Airflow monitoring system repeatability shall be to within ± 1.5 percent of full-scale.

2. Airflow sensor.

a. Required operating principle. The sensor shall employ a thermal anemometer technique for sensing airflow.

b. Physical characteristics. The sensor shall consist of a cylindrical probe and an attached/integral electronics processing package. The sensor probe shall be constructed of corrosion resistant material. The sensor probe shall be from 1.0 cm to 2.6 cm in outside diameter. The probe length shall be appropriate for the cylindrical or rectangular duct size as specified. A mounting plate shall be provided to secure the sensor to the air duct. The sensor shall be readily removable from the mounting plate (to facilitate periodic cleaning).

c. Power input. The airflow sensor shall be powered by +15 to +18 Vdc and shall draw no more than 350 mA.

d. Airflow Range. The airflow velocity range of the sensor shall be from zero mps to the full-scale mps value as specified.

e. Output signal. The airflow sensor shall output a 4-20 mA signal linearly proportional to airflow velocity. A 4 mA output shall correspond to zero airflow velocity and a 20 mA output shall correspond to full-scale airflow velocity. The sensor shall drive a resistive load of 250 ohms ± 5 percent.

f. Accuracy. The airflow velocity output signal shall be accurate to within ± (3.0 percent of reading plus 1 percent of full-scale).

g. Repeatability. Repeatability shall be to within ± 0.5 percent of full-scale.

h. Response time, velocity change. When operating in a standard one-atmosphere environment and an airflow velocity of 25 percent full-scale; the airflow sensor shall achieve a full-accuracy output signal within 10 seconds for an airflow velocity step increase to 75 percent full-scale.

i. Response time, temperature change. When operated in a standard one-atmosphere environment and an airflow velocity of 50 percent full-scale; the airflow sensor shall achieve a full-accuracy output signal within four minutes for a 20 degrees Celsius (°C) temperature step increase.

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j. Interface with airflow display panel. The airflow sensor shall connect to the airflow display panel via a three-conductor cable of a length as specified. One conductor shall carry the airflow signal, one conductor the dc power, and one shall be a common conductor. The cable shall be detachable from the airflow sensor via an accessible screw-lug terminal located inside the airflow sensor assembly. The screw-lugs shall be marked as to function.

k. Reliability. The airflow sensor minimum mean time between failures (MTBF) shall be no less than 25,000 hours.

l. Safety (hazardous environment). The airflow sensor shall be in compliance with the NEC requirements for safe use in Class I, Division 2, Group D hazardous locations.

3. Airflow display panel.

a. Physical Characteristics. The airflow display panel shall not exceed the following outside dimensions: 33 cm in width, 28 cm in height, and 25.4 cm in depth (including any attached mounting pads). The panel shall be bulkhead mountable via the hole pattern as specified. (An adapter plate for matching the panel to the specified hole-pattern may be used.) The panel shall be corrosion resistant. The panel shall incorporate a swing-open panel door with an identification plate affixed on the outside of the panel door and labeled as indicated.

b. Range. The airflow range of the panel shall be manually selectable. Selectable ranges shall include:

0 - 1	mps	0 - 10	mps
0 - 2.5	mps	0 - 25	mps
0 - 5	mps	0 - 50	mps

c. Input signal. The airflow panel shall accept a 4-20 mA signal linearly proportional to airflow velocity. A 4 mA output shall correspond to zero airflow velocity and a 20 mA output shall correspond to full-scale airflow velocity. Input impedence for the signal shall be resistive at 250 ohms ± 1 percent.

d. Airflow display. The panel shall incorporate an illuminated digital readout. The digital readout shall display airflow velocity or trip point setting in meter-per-second (mps) on a selectable basis. Airflow velocity displayed shall be mean airflow velocity for the most recent one-second interval.

e. Trip-point set. The panel shall enable the setting of a low-airflow alarm trip-point for any value in the selected airflow range. The panel shall detect whenever airflow velocity drops below the trip-point setting for a period of 30 ± 5 seconds and shall set a low-airflow alarm state accordingly. The panel shall retain memory of the trip-point setting at all times, including power loss times.

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f. Audible alarm. The panel shall incorporate an audible alarm which generates an audio output of from 75 to 95 dB at two feet. The audible alarm shall sound whenever a low-airflow alarm state is first set and shall remain activated until: 1) airflow rate rises above the trip-point setting for at least 10 seconds, or 2) an audible alarm silence switch that shall be incorporated in the panel is manually operated. The audible alarm silence switch shall be labeled as to function.

g. Visual alarm. The panel shall incorporate a visual alarm which flashes on and off when an alarm state is set. Alarm flashing shall not cease until the alarm state is reset to indicate normal airflow. The visual alarm shall be labeled as to function.

h. Test Mode. The panel shall incorporate a manually-initiated test mode which demonstrates/verifies functionality of the visual and audible alarms.

i. Airflow velocity output signal. The panel shall output a 0 Vdc to 10 Vdc analog scalar signal linearly proportional to airflow velocity over the specified range. The output circuitry shall supply a 10 mA maximum current and shall be "short" protected. The airflow velocity output signal shall be instantaneous airflow velocity damped by a one-second time constant. The airflow velocity signal shall be available on an accessible screw-lug terminal located inside the airflow panel. The screw-lugs shall be marked as to function.

j. Airflow-loss alarm output signals. The panel shall provide two isolated switch-closure discrete outputs. Closed-contacts shall indicate a low-airflow alarm state or that power has been lost for more than 30 ± 5 seconds. Open-contacts output shall indicate that airflow and power are normal. The switch contacts shall be rated at not less than 1.5 Amps, 115 Vac. The switch-closure output signals shall be made available on an accessible screw-lug terminal located inside the airflow panel. The screw-lugs shall be marked as to function.

k. Accuracy. The airflow panel circuitry shall add no more than ± 1.75 percent error to the data received from the airflow sensor. The ± 1.75 percent maximum allowable error applies to both the digital display of airflow velocity and to the Vdc analog scalar output signal which carries airflow velocity information. Panel trip-point detection accuracy shall be within ± 3.0 percent of the nominal trip-point setting.

l. Response time. When operating in a standard one-atmosphere environment and an airflow velocity of 25 percent full-scale; the airflow display panel shall achieve a full-accuracy output signal within 1.1 seconds for an airflow velocity signal step increase to 75 percent full-scale. For the same conditions, the airflow velocity display shall achieve a full-accuracy readout of airflow velocity within 2 seconds.

m. Repeatability. Repeatability shall be to within ± 1.0 percent of full-scale.

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n. Sensor fault detect. The panel shall incorporate a sensor fault indicator. The panel shall determine whenever the airflow sensor input signal strays out of the nominal 2 mA to 20 mA operating range and shall generate a sensor fault alarm accordingly.

o. Power. The airflow display panel shall be nominally powered by 115 volt 60 Hz power. An ac input power on/off switch shall be provided. Input ac power shall be fused.

p. Power output. The airflow display panel shall output +15 to +18 Vdc power, 350 mA maximum, for the airflow sensor.

q. Interface with airflow sensor. The panel shall connect to the airflow sensor via a three-conductor cable. One conductor shall carry the airflow signal, one conductor the dc power, and one shall be a common conductor. The cable shall be detachable from the panel sensor via an accessible screw-lug terminal located inside the airflow sensor assembly. The screw-lugs shall be marked as to function.

r. Reliability. The airflow display panel minimum mean time between failures (MTBF) shall be no less than 28,000 hours.

s. Enclosure. The airflow display panel as mounted in the normal position shall be dripproof. Liquid droplets or solid particles falling on the equipment from the vertical (± 15 degrees) shall cause no harmful operational/ functional effects.

4. General requirements.

a. Accessibility. All parts and subassemblies which may require servicing, repair, or replacement during the life of the airflow sensing equipment shall be readily accessible.

b. Interchangeability. Airflow sensors built in accordance with this CID shall be interchangeable for the specified airflow range. Airflow display panels built in accordance with this CID shall be interchangeable.

c. Vibration. The airflow monitoring system equipments shall be in compliance with the vibration tolerance requirements of MIL-STD-167-1 for Type I equipment.

d. Shock. The airflow monitoring system equipments shall be in compliance with MIL-S-901 Grade (B), Class I, Light Weight, Type B mechanical shock tolerance standards.

e. Temperature. The airflow monitoring system equipments shall meet the stated accuracy requirements over an operational temperature range of from 0°C to +50°C.

f. EMI. The airflow monitoring system equipments shall be in compliance with the Federal Communications Commission (FCC) Class B Rules for computing devices.

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Quality assurance provisions.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

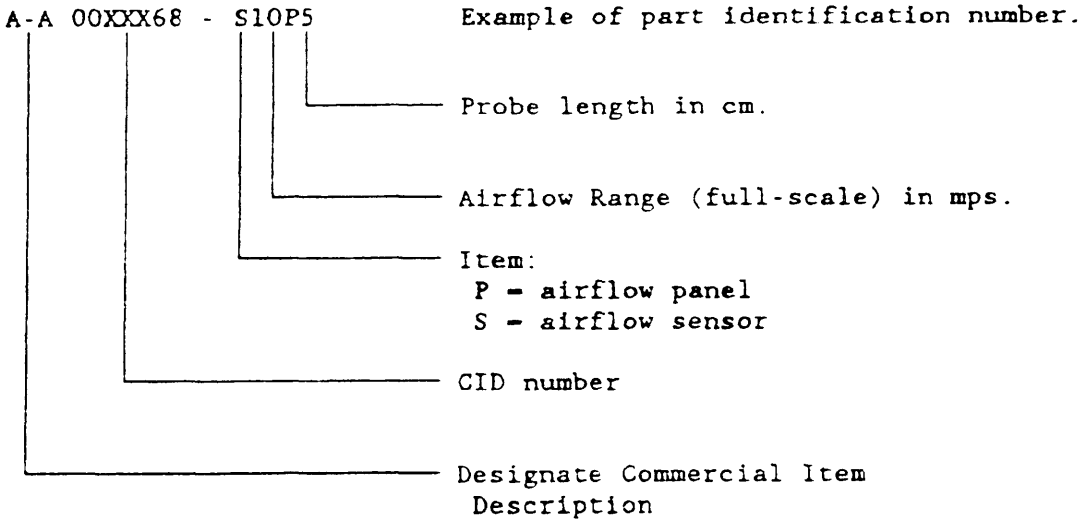
The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking.

Preservation, packaging, packing, labeling, and marking shall be as specified.

Identification Numbers.

The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



Notes.

1. Ordering data. Acquisition documents should specify the following:

   a. Specify unit type.

      P - airflow panel and interface cable

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S = airflow sensor and mounting plate

b. Specify sensor full-scale (that is, maximum expected) airflow rate value in meters-per-second (mps).

0 - 1	mps	0 - 10	mps
0 - 2.5	mps	0 - 25	mps
0 - 5	mps	0 - 50	mps

c. Specify air duct cross-sectional dimensions:

1) for cylindrical duct: diameter in meters or centimeters
2) for rectangular/square duct: length x width in meters or centimeters

d. Airflow display panel identification plate labeling. The airflow display panel identification plate shall be labeled "AIRFLOW PANEL" and shall identify the applicable CID number, manufacturer name, and part number of the airflow display panel.

e. Airflow panel mounting hole pattern. Specify the following hole pattern for naval ship procurements: three 1.55-cm diameter holes configured to form a 30.5 x 25.4 x 25.4 cm isosceles triangle with the longest side in the horizontal plane at the top of the airflow display panel.

f. Specify sensor-panel interface cable length in meters.

MILITARY INTERESTS:

Custodians
Army - ME
Navy - SH
Air Force - 99

Review activities
Army - ER
Navy - AS
Air Force - 13

User activities
Navy - MC
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:
COMMERCE - NITS
GSA-FSS

PREPARING ACTIVITY:
NAVY - SH
(Project 6680-0233)

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